

Object Groups:

- o In large networks especially Data Centers; the ACLs can be too big.
- o Up to hundreds of lines and difficult to configure and manage ACLs.
- o Object group-based Access Control Lists provide the simple solution.
- o Object group is smaller, readable, and easier to configure and manage.
- o Object Groups allows classifying users, devices, or protocols into groups.
- o Then object groups can be applied to the Access Control Lists (ACLs).
- o Single ACE allow an entire group of users to access a group of user.
- o Single ACE allow an entire group of users to access group of server groups.
- o An object represents any single item in the network environment.
- o Also, any changes in hosts, ports etc are done inside the object-group.
- o The changes are automatically reflected in the Access Control List (ACL).
- o Two types of objects can be configured network object & service object.
- o The Cisco ASA Firewall version 9.x supports Six (6) types of object group.

ASA1(config)# **object-group ?**

configure mode commands/options:

icmp-type	Specifies a group of ICMP types, such as echo
network	Specifies a group of host or subnet IP addresses
protocol	Specifies a group of protocols, such as TCP, etc
security	Specifies identity attributes such as security-group
service	Specifies a group of TCP/UDP ports/services
user	Specifies single user, local or import user group

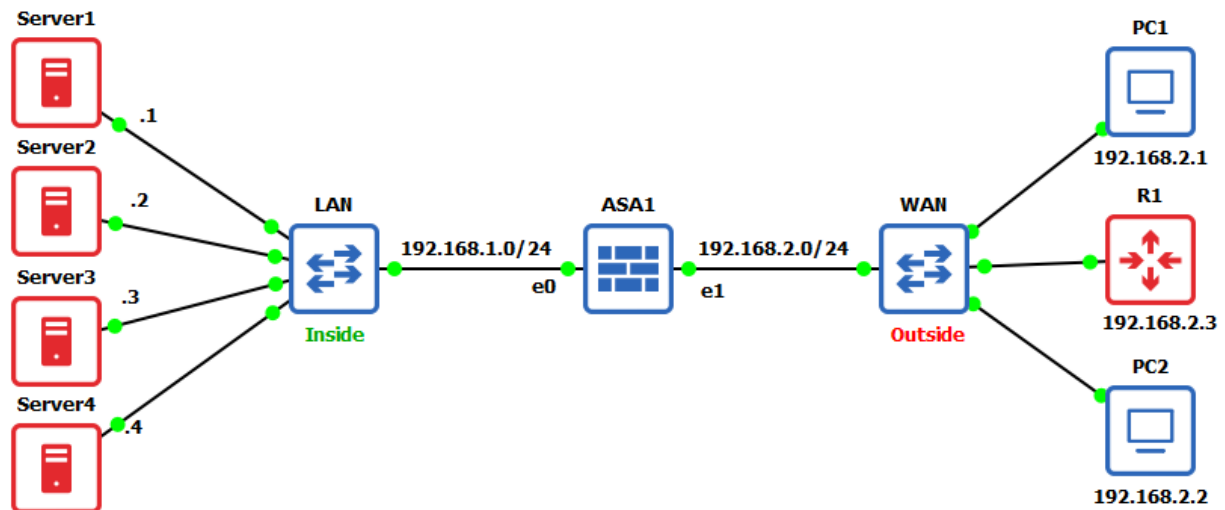
Network	Used to group together hosts or subnets.
Service	Used to group TCP or UDP port numbers.
Protocol	Used to group protocols TCP, UDP, GRE, ESP, AH, OSPF etc.
ICMP-Type	Used to group ICMP message types. Example echo, echo-reply etc.
User	Creates Local User Groups and Imported Active Directory User.
Security	Object group Used with Cisco TrustSec.

ASA1(config-network-object-group)# **object-group network net**
ASA1(config-network-object-group)# ?

description	Specify description text
group-object	Configure an object group as an object
help	Help for network object-group configuration commands
network-object	Configure a network object
no	Remove an object or description from object-group

ASA1(config-network-object-group)#

Lab Time:



Server1 Toolbox Docker	# Static config for eth0 auto eth0 iface eth0 inet static address 192.168.1.1 netmask 255.255.255.0 gateway 192.168.1.100 up echo nameserver 8.8.8.8 > /etc/resolv.conf
Server2 Toolbox Docker	# Static config for eth0 auto eth0 iface eth0 inet static address 192.168.1.2 netmask 255.255.255.0 gateway 192.168.1.100 up echo nameserver 8.8.8.8 > /etc/resolv.conf
Server3 Toolbox Docker	# Static config for eth0 auto eth0 iface eth0 inet static address 192.168.1.3 netmask 255.255.255.0 gateway 192.168.1.100 up echo nameserver 8.8.8.8 > /etc/resolv.conf

Server4 Toolbox Docker	<pre># Static config for eth0 auto eth0 iface eth0 inet static address 192.168.1.4 netmask 255.255.255.0 gateway 192.168.1.100 up echo nameserver 8.8.8.8 > /etc/resolv.conf</pre>
PC1 Webterm Docker	<pre># Static config for eth0 auto eth0 iface eth0 inet static address 192.168.2.1 netmask 255.255.255.0 gateway 192.168.2.100 up echo nameserver 8.8.8.8 > /etc/resolv.conf</pre>
PC2 Webterm Docker	<pre># Static config for eth0 auto eth0 iface eth0 inet static address 192.168.2.2 netmask 255.255.255.0 gateway 192.168.2.100 up echo nameserver 8.8.8.8 > /etc/resolv.conf</pre>
R1 Configuration	<pre>R1(config)#interface Ethernet0/0 R1(config-if)#ip address 192.168.2.3 255.255.255.0 R1(config-if)#no shutdown R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.2.100</pre>
Cisco ASA Firewall Configuration	<pre>ciscoasa# config t ciscoasa(config)# hostname ASA1 ASA1(config)# interface e0 ASA1(config-if)# ip address 192.168.1.100 ASA1(config-if)# nameif inside ASA1(config-if)# no shutdown ASA1(config-if)# exit ASA1(config)# interface e1 ASA1(config-if)# ip address 192.168.2.100 ASA1(config-if)# nameif outside ASA1(config-if)# no shutdown</pre>

Access List without Object Group

```
ASA1(config)# access-list SERVER-Access permit tcp any host 192.168.1.1 eq 80
ASA1(config)# access-list SERVER-Access permit tcp any host 192.168.1.2 eq 80
ASA1(config)# access-list SERVER-Access permit tcp any host 192.168.1.3 eq 80
ASA1(config)# access-list SERVER-Access permit tcp any host 192.168.1.4 eq 80
ASA1(config)# access-group SERVER-Access in interface outside
```

Access List with Object Group

```
ASA1(config)# clear configure access-list SERVER-Access
ASA1(config)# object-group network SERVERS
ASA1(config-network-object-group)# network-object host 192.168.1.1
ASA1(config-network-object-group)# network-object host 192.168.1.2
ASA1(config-network-object-group)# network-object host 192.168.1.3
ASA1(config-network-object-group)# network-object host 192.168.1.4
ASA1(config)# access-list SERVER-Access extended permit tcp any object-group SERVERS eq 80
ASA1(config)# access-group SERVER-Access in interface outside
```

Access List with Service Object

```
ASA1(config)# object-group service SERVER-Port tcp
ASA1(config-service-object-group)# port-object eq ftp
ASA1(config-service-object-group)# port-object eq www
ASA1(config)# access-list SERVER-Access extended permit tcp any object-group SERVERS object-group SERVER-Port
ASA1(config)# access-group SERVER-Access in interface outside
```

ASA1# show access-list	ASA1# show running-config access-list
ASA1# show running-config access-group	ASA1# show object-group

```
ASA1# show access-list
access-list cached ACL log flows: total 0, denied 0 (deny-flow-max 4096)
      alert-interval 300
access-list SERVER-Access; 8 elements; name hash: 0xaa74ff3a
access-list SERVER-Access line 1 extended permit tcp any object-group SERVERS object-group SERVER-Port
4a43
access-list SERVER-Access line 1 extended permit tcp any host 192.168.1.1 eq ftp (hitcnt=4)
access-list SERVER-Access line 1 extended permit tcp any host 192.168.1.1 eq www (hitcnt=3)
access-list SERVER-Access line 1 extended permit tcp any host 192.168.1.2 eq ftp (hitcnt=0)
access-list SERVER-Access line 1 extended permit tcp any host 192.168.1.2 eq www (hitcnt=5)
access-list SERVER-Access line 1 extended permit tcp any host 192.168.1.3 eq ftp (hitcnt=0)
access-list SERVER-Access line 1 extended permit tcp any host 192.168.1.3 eq www (hitcnt=0)
access-list SERVER-Access line 1 extended permit tcp any host 192.168.1.4 eq ftp (hitcnt=0)
access-list SERVER-Access line 1 extended permit tcp any host 192.168.1.4 eq www (hitcnt=0)
```